

Predicting Education-Job Mismatch: Field-Dependent Institution Signaling with Evidence from LinkedIn Data

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August 14, 2026

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Abstract

This work addresses the measurement and prediction of education-job mismatch, aggregating both horizontal and vertical dimensions, within the Argentine labor market. Leveraging educational and professional data from LinkedIn profiles, we implement a novel classification framework that integrates manual labeling with Large Language Model (LLM) assistance to empirically identify mismatches. The predictive exercise in this paper evaluates the moderating role of institutional prestige—defined by market-driven perceptions—across a selected subset of Argentine universities. By focusing on two contrasting disciplines, Engineering and Psychology, we examine the alignment between our empirical findings and established institutional hierarchies, such as the QS World University Rankings. This paper contributes to the literature by providing a granular assessment of how university reputation and field choice interact to shape professional trajectories.

Keywords: Education-Job Mismatch, Machine Learning, Classification, Signaling, Institutional Prestige.

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1 Introduction

Human capital accumulation is widely known as a fundamental factor for economic growth and social mobility (Becker, 1975). Analyzed as an individual investment choice, the efficiency of investing in education depends critically not only on the skill acquisition, but also on the subsequent allocation within the labor market. A misalignment between the knowledge acquisition (education) and the knowledge utilization (productive labor) represents a resources misallocation: graduates who undertook costly educational investments—in both time and pecuniary terms—find that their acquired human capital is not demanded by the market.

The existing literature consensus posits that horizontal mismatch —the discrepancy between an individual’s field of study and their occupational field— constitutes a significant labor market friction, generally entailing wage penalties, lower occupational status, and lower job satisfaction (Somers et al., 2019). In the United States, approximately 20% of graduates report their job to be entirely unrelated to their degree, facing a wage penalty estimated between 3% and 4% compared to their well-matched peers (Robst, 2007; Rios-Avila and Saavedra-Caballero, 2019). While some evidence finds that horizontal mismatch only penalize wages when combined with vertical mismatch —the discrepancy in the person’s level of education and the one required by the job- (Mahuteau et al., 2015), the consensus remains that the underutilization of field-specific human capital represents a loss in productive efficiency. This paper explores this phenomenon by utilizing a unified mismatch indicator that aggregates both horizontal and vertical dimensions into a single measure of labor misalignment.

Furthermore, the persistence of these mismatches is often amplified by the business cycle; for instance, cohorts entering the labor market during recessions exhibit higher probabilities of long-term displacement (Pérez Navarro, 2021). In the context of emerging economies like Argentina, the study of educational mismatch acquires a distinct dimension due to persistent macroeconomic instability and structural labor market characteristics. These environments often feature significant information asymmetries that lead to sub-optimal educational choices. For instance, evidence from the Chilean market (Fiscalía Nacional Económica, 2025) suggests that applicants frequently overestimate vocational factors while undervaluing critical economic indicators such as employability and expected earnings—a pattern likely mirrored in the Argentine context.

Following the signaling microeconomic theory proposed by Spence (1978), we argue that these misallocations are closely related to the imperfect information surrounding educational decisions. Publicly available data on universities often omit graduate placement statistics, making it more difficult for potential students to quantify the "investment return". This lack of data ignores a critical reality: variance in labor outcomes is often greater between institutions teaching the same discipline than between different sub-

fields within a single institution. The lack of detailed information about the university-differentiated value of higher education act as an exacerbating factor of misallocation, turning the decision of which field of study to pursue into a risky bet. If there exists heterogeneity in the strength of the skill signal depending on the educational institution attended, the probability of mismatch in the labor market is not uniform across universities, even when considering the same degree. Since it positively associates with the quality of knowledge transmission, the prestige of a university acts as a signaling mechanism that reduces frictions (mismatches) in their graduates' labor-market performance. Providing more information in this aspect represents one of the core contributions of this paper.

Nevertheless, the magnitude of these effects is also heterogeneous across disciplines, appearing highly sensitive to the specificity of the skills acquired. Graduates from "generalist" majors, such as Arts and Social Sciences, often exhibit a higher risk of mismatch but lower relative penalties, whereas those in "specialist" fields like Health Sciences or Engineering encounter lower frequencies of mismatch but suffer higher wage losses when they occur (Pérez Navarro, 2021; Robst, 2007). Considering this, we examine education-job mismatch over two large and contrasting disciplines usually chosen by Argentine undergraduates: Psychology, as the major providing general skills, and Engineering, as the specific-skill major. To further simplify the study while maintaining competitive programs in the analysis, we confine the set of Argentine universities studied to those listed in the 2026 QS Ranking and examine their graduates' observed matching behavior in the two disciplines considered. Then, we contrast our results with the mentioned ranking, evaluating the skill signal granted by universities for these two specific fields.

Despite the richness of the global evidence, empirical applications in emerging economies like Argentina remain scarce, particularly those capable of capturing dynamics through granular, non-traditional data sources. Indeed, the empirical identification of education-job mismatch remains a subject of methodological debate, primarily centered on the tension between subjective and objective measurement approaches. Somers et al. (2019) systematically reviews the literature published between 1995 and 2015, highlighting a persistent lack of uniformity in definitions that leads to widely varying incidence rates. For instance, while Robst (2007) relies on subjective, self-reported measures of degree relevance, Rios-Avila and Saavedra-Caballero (2019) propose objective indices based on the observed distribution of workers across occupations in the United States. Conversely, Pérez Navarro (2021) utilize a dual approach, incorporating both self-perception and objective indicators to capture overqualification. In a more related approach, Mahuteau et al. (2015) measures horizontal mismatch based on the mode of the field of study within an occupation. This paper adopts an objective empirical strategy to assess the mismatch, using granular and high-dimensional educational and professional data from LinkedIn profiles. We identify mismatches by evaluating the alignment between a graduate's professional role and their field of study, employing both manual labeling and Large Language

Model (LLM) assistance to construct a unified education-job mismatch indicator.

In the regional context, there exists recent antecedents of non-supervised learning algorithms application in analyzing the human capital return. Galvis Caballero (2021) implement ensemble models to predict income and monetary poverty in Colombia, and shows that these algorithms not only significantly reduce out-of-sample mean squared error (MSE) relative to linear specifications but also—through the application of SHapley Additive exPlanations (SHAP) values—reveal the complex, non-linear structure of returns to education and experience. This approach is particularly relevant to our proposal, as it suggests that the educational 'signal' interacts with the labor market in a heterogeneous and non-linear fashion.

Rather than evaluating the effects of education and profession misalignment, we focus on predicting mismatches and using that prediction to assess the heterogeneity of universities skill-signals. While traditional works like Robst (2007) use ordered logistic regressions for the probability of a mismatch, we rely on Machine Learning techniques to predict an education-job match. As a preliminary step, we first estimate a Least Absolute Shrinkage and Selection Operator (LASSO) model. This traditional regularization technique allows us to establish a linear benchmark and identify the most statistically relevant universities within our sample by performing automated variable selection. Subsequently, we transition toward Gradient Boosting frameworks, which are better suited to capture the non-linearities, and contrast the results.

While traditional econometric methods often rely on rigid distributional assumptions, our machine learning strategy exploit observed labor market outcomes to detect the specific matching patterns between educational backgrounds—institution and field of study—and professional occupations. This framework further enables an examination of how institutional prestige moderates the probability of following a mismatched professional trajectory. Ultimately, we provide predictive estimates of matching based on the empirical probability of effective entry into corresponding segments of the Argentine labor market. In doing so, this paper offers novel evidence on education-job mismatch, disaggregated by both educational institution and disciplinary field.

2 Data

2.1 Data Extraction

Given the large number of universities (111) and degree programs offered in Argentina, a subset of the most prominent institutions was selected for this analysis. The selection was not arbitrary: only universities included in the prestigious *QS Quacquarelli Symonds* ranking were considered. Specifically, the sample comprises the 18 Argentine universities listed in the 2026 ranking ([QS World University Rankings, 2026](#)). Two degrees were chosen for the purpose of this analysis: Engineering and Psychology. These disciplines were selected due to their contrasting labor market pathways and skill sets: Engineering typically leads to highly technical, structured career trajectories, whereas Psychology graduates often pursue more flexible and service-oriented roles. This contrast allows the analysis to capture how university quality and reputation may differently impact graduates' labor market outcomes across diverse types of programs.

Individual-level data were entirely extracted from *LinkedIn*. This platform provides an ideal source for the purpose of this study, as both educational and career trajectories are publicly available. However, it also posed a significant challenge: data scrapping is extremely difficult given the strong security that *LinkedIn* employs against such practices, meaning that all information had to be collected manually. To perform this task, avoiding *LinkedIn* biased selection was necessary. Random profiles were selected from the alumni listing of each university, applying a filter with the fields of study (Engineering and Psychology) as keywords. Randomization was ensured by generating a random number from the the first 100 profiles of the listing. If the resulting number selected a current student or someone who had a different affiliation with the university (a master's degree instead of a bachelor's, for example), the number was sorted again until the candidate met the required characteristics. Following this method, 16 profiles were extracted for each university; 8 in Engieneering and 8 in Psychology. Some universities, however, did not have available data for one¹ or both² of these fields, and were therefore omitted from the analysis. Particular data included each profile's genre, graduation date, university, degree, general study area and the first job performed after graduation.

2.2 Matching assignment

To simplify the analytical framework and capture the overall efficiency of human capital allocation, we aggregate the two dimensions into a binary composite indicator, as explained in the previous section. Under this definition, the condition of 'mismatch' is satisfied if a graduate experiences either horizontal (field-specific) or vertical (qualification-

¹UNCUYO, UNICEN , UNL, UTN, UNSAM, UNS

²UTDT, UDESA.

level) misalignment. Formally, we define the indicator $Match_i$ such that $Match_i = 1$ if $H_i = 1 \vee V_i = 1$, and 0 otherwise.³

Three different approaches are tested to evaluate the match between each profile’s degree and the first job held after graduation. First, we perform a manual classification, given that the sample consists of only 222 observations, following the criteria explained above. Concerned that this approach might be biased, we run two additional simulations using OpenAI’s API in Python. This approach is advantageous because the models apply a consistent and standardized criterion across all observations, reducing potential inconsistencies that may arise from manual evaluation. Specifically, we ask GPT-4.1 and GPT-4o mini to perform the matching task using the prompt displayed in Appendix ??.

Once this process is completed, the dataset contains a match indicator for each observation under all three classification approaches, leaving the data ready for the subsequent empirical analysis. The prompt was formulated in Spanish, as degree titles and job positions are originally reported in Spanish, ensuring a more accurate interpretation of the information.

³Where H and V represents horizontal and vertical mismatch respectively.

3 Empirical Strategy and Methodology

Our methodological approach employs a two-stage machine learning sequential estimation designed to balance interpretability with predictive accuracy. Given the strong limitation in our sample size ($N = 288$), we adopt a "sparse-to-dense" strategy to mitigate the risks of overfitting while uncovering non-linear professional trajectories. The first stage utilizes *Least Absolute Shrinkage and Selection Operator* (LASSO) regularization for automated feature selection, while the second stage leverages *Extreme Gradient Boosting* (XGBoost) to capture complex, non-linear patterns in the data. This sequential framework allows us to identify the most relevant predictors while maintaining sufficient model flexibility to detect institutional and field-specific heterogeneity in matching probabilities.

3.1 First Stage: Feature Selection via LASSO Regularization

In the initial stage, we implement a LASSO model within a penalized logistic regression framework. To address the high dimensionality in our dataset—stemming from the extensive set of categorical dummies for specific universities—we apply an L_1 norm penalty to the objective function:

$$\min_{\beta} \left\{ \sum_{i=1}^n \mathcal{L}(y_i, f(x_i)) + \lambda \sum_{j=1}^p |\beta_j| \right\}$$

where x_i is the vector of covariates (including university dummies, field of study, graduation year, and genre), $\mathcal{L}$ represents the logistic loss, and λ is the regularization hyperparameter determined via 5-fold cross-validation over a logarithmically-spaced grid ranging from 10^{-2} to 10^2 (50 points), maximizing the Area Under the ROC Curve (AUC-ROC). All features are standardized to have mean zero and unit variance using `StandardScaler` to ensure comparability of penalty magnitudes across variables with different scales. The data is partitioned into training (80%) and test (20%) sets using stratified sampling to preserve the distribution of the outcome variable across splits.

The primary objective of this stage is relevance identification. By inducing sparsity, the LASSO algorithm effectively filters the feature space, shrinking the coefficients of variables with negligible marginal contributions to zero.⁴ This property makes Lasso particularly suitable for high-dimensional settings where the number of potential predictors (in our case, numerous university dummies) may approach or exceed the number of observations, or where multicollinearity is a concern. From a theoretical point of view, we argue that this property can be thought as an examination of market signals: universities whose coefficients are discarded are interpreted as institutions whose "skill signal" lacks sufficient

⁴We retain only variables with non-zero coefficients above a threshold of $|\hat{\beta}_j| > 0.01$, which is more stringent than the conventional choice of 0.001.

statistical salience to predict labor market alignment within a linear framework. This step ensures that the subsequent model is trained on a refined, signal-dense subset of attributes.

3.2 Second Stage: Non-Linear Classification via XGBoost

Using exclusively the variables selected by Lasso (denoted as $\mathcal{S} = \{j : |\hat{\beta}_j^{\text{Lasso}}| > 0.01\}$), we train an XGBoost classifier. While the first stage identifies linear relevance, the XGBoost architecture is deployed to capture the complex, non-linear interactions that characterize professional outcomes. This model is a highly efficient implementation of gradient boosted decision trees that operates as an ensemble technique, constructing an additive ensemble of decision trees through sequential boosting:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), \quad f_k \in \mathcal{F} \quad (1)$$

where $\mathcal{F}$ denotes the space of regression trees, and each tree f_k is learned sequentially to minimize the residual loss of the preceding ensemble.

Unlike linear specifications, the additive structure allows XGBoost to approximate complex, non-linear decision boundaries and detect idiosyncratic interactions, such as the specific premium or penalty associated with a "Degree X" when granted by "University Y." For instance, the signaling value of a particular university may vary non-monotonically across fields of study or graduation cohorts, patterns that linear models cannot detect. Moreover, XGBoost incorporates internal L_1 and L_2 regularization, which, combined with our initial LASSO filter, provides a dual-layer defense against the likely presence of noise.

We employ `GridSearchCV` with 5-fold cross-validation to select optimal hyperparameters over a comprehensive grid. Maximum tree depth ($d_{\max}$) is varied across $\{3, 5, 7\}$ to control model complexity and the maximum order of interaction effects that can be captured. The learning rate (η) is tested at $\{0.01, 0.1, 0.3\}$, determining the contribution of each tree to the ensemble; lower values require more boosting iterations but reduce overfitting risk. The number of estimators (K) varies across $\{50, 100, 200\}$, representing the total number of boosting iterations. Subsample ratio is set to $\{0.8, 1.0\}$, where values below unity introduce stochasticity by using only a random fraction of training data per tree, thereby reducing variance. Finally, column subsample ratio varies across $\{0.8, 1.0\}$, controlling the fraction of features considered per split to promote diversity among trees and prevent over-reliance on dominant predictors. The model selection criterion is AUC-ROC under cross-validation, consistent with the First Stage. This metric is particularly appropriate for binary classification problems with modest imbalance, as it evaluates discriminatory power across all classification thresholds rather than at a single arbitrary cutoff point.

The decision to implement a sequential *LASSO-XGBoost* pipeline—rather than de-

ploying the boosting algorithm directly on the full feature set—is a strategic choice to "optimize" the bias-variance trade-off. Although XGBoost is inherently robust, the high cardinality of our categorical variables relative to our limited sample size ($N = 288$) poses a significant risk of overfitting to idiosyncratic noise. By utilizing LASSO as a preliminary filter, we reduce the model’s complexity and variance, ensuring that the final ensemble focuses on robust, generalizable patterns. This hybrid approach results in a final model that achieves high predictive accuracy while maintaining the structural integrity required for economic analysis.

We execute this evaluation protocol separately for each of the three classification schemes (Manual, GPT-4.1, and GPT-4o) to assess consistency across labeling approaches. High concordance in predictive performance across schemes would suggest that the underlying signal is robust to measurement differences. Additionally, we compare Lasso and XGBoost performance on the same test set to quantify the incremental predictive gain from allowing non-linearities and interactions. This comparison directly tests the hypothesis that labor market matching exhibits complex patterns not captured by linear models.

4 Results

The sequential Lasso-XGBoost framework exploits complementary strengths of both methods while addressing their respective limitations. We also report multiple complementary performance metrics to provide a comprehensive assessment of model quality. The primary metric is AUC-ROC (Area Under the Receiver Operating Characteristic Curve). Ranging from 0.5 (random guessing) to 1.0 (perfect classification), AUC measures the model’s ability to discriminate between matched and mismatched graduates across all decision thresholds. Accuracy, defined as the proportion of correctly classified observations, provides an intuitive measure of overall model performance but can be misleading under class imbalance. Therefore, we prioritize AUC-ROC for model comparison while reporting accuracy for completeness.

Table 1: Lasso Feature Selection Results Across Classification Schemes

Metric	GPT-4.1	GPT-4o	Manual
<i>Panel A: Sample Composition</i>			
Training observations	176	176	176
Test observations	45	45	45
<i>Panel B: Hyperparameter Selection</i>			
Optimal λ (CV)	0.0100	2.3300	10.4811
<i>Panel C: Variable Selection</i>			
Total variables	18	18	18
Selected variables	17	14	5
Discarded variables	1	4	13
Selection rate (%)	94.4	77.8	27.8
<i>Panel D: Test Set Performance</i>			
AUC-ROC	0.6096	0.7109	0.7510
Accuracy	0.6000	0.6667	0.6222
MSE	0.4000	0.3333	0.3778

The MSE values (ranging from 0.33 to 0.40) confirm that the GPT-4o scheme provides the lowest average error in this specific exercise. However, the superior AUC-ROC of the Manual labels justifies its use as the "ground-truth" for establishing the most reliable institutional signals. This comparison highlights why your hybrid approach is necessary: using LLMs allows for scalability, but the manual calibration is what ensures the model is anchored to high-quality economic signals.

By shrinking less informative coefficients to zero, LASSO provides an interpretable mapping of institutional effects on mismatch probabilities. In this sense, Lasso coefficients offer preliminary insights into the direction and magnitude of institutional effects, facilitating economic interpretation. The sign of the Lasso coefficient indicates whether a particular university is associated with higher or lower match probabilities relative to the baseline category, while the magnitude reflects the strength of this association under the linear approximation. This can be consulted in the [A5](#) available in the Appendix.

The signaling value of a particular university may vary non-monotonically across fields of study or graduation cohorts, exhibit threshold effects, or interact with labor market conditions in ways that violate the linear-additive structure imposed by Lasso. XGBoost, in turn, captures non-linearities and higher-order interactions that linear models cannot accommodate. By restricting XGBoost to the Lasso-selected feature space, we prevent the model from learning spurious patterns from noise variables while retaining sufficient flexibility to detect genuine non-linear relationships. Beyond binary classification (match

vs. mismatch), we take advantage of the XGBoost model and estimate the conditional probability of success. This allows for a continuous ranking of educational institutions based on their empirical capacity to generate effective signals that culminate in a first-job match.

To interpret the XGBoost model’s predictions and identify the most influential determinants of labor market matching, we compute feature importance scores using the gain-based metric. Importantly, feature importance reflects predictive power rather than causal effects. A high-importance variable is one that substantially reduces prediction error when used for splitting decisions, regardless of whether it increases or decreases match probability.⁵ The following table reports predicted match probabilities (in percentage points) for university-field combinations across the three classification schemes.

Table 2: Predicted Match Probabilities by University-Field Combination Across Classification Schemes

University	Manual		GPT-4o		GPT-4.1	
	Eng.	Psych.	Eng.	Psych.	Eng.	Psych.
<i>Panel A: Top-Ranked Private Universities</i>						
Universidad Argentina de la Empresa	55.73	52.18	–	–	71.00	75.35
Universidad Austral	–	–	–	–	68.32	72.92
Universidad de Belgrano	52.54	50.53	54.58	46.86	54.96	60.37
<i>Panel B: Elite National Universities</i>						
Universidad Nacional de La Plata	–	–	–	–	70.75	75.12
Universidad Nacional de Córdoba	–	–	77.23	70.58	–	–
Universidad Nacional de Rosario	–	–	76.35	72.14	–	–
<i>Panel C: Regional National Universities</i>						
Universidad Nacional del Litoral	–	–	59.25	53.53	–	–
Universidad Nacional de San Luis	–	–	47.06	41.35	27.90	34.68
<i>Panel D: Other Private Institutions</i>						
Universidad Católica Argentina (UCA)	–	–	39.31	30.92	59.26	67.14
Universidad de Palermo	–	–	46.49	35.81	27.78	27.27
<i>Panel E: Summary Statistics</i>						
Mean predicted probability	54.15	51.36	57.22	54.36	54.46	61.56
Standard deviation	1.60	0.83	13.82	14.96	18.34	19.71

Notes: Predictions are generated using the XGBoost model trained on each scheme’s labels, holding control variables (graduation year, gender, first job) at median/modal values. Entries marked with “–” indicate universities that were either not selected by Lasso (coefficient forced to zero) or exhibited zero feature importance in XGBoost, meaning they are statistically indistinguishable from the baseline category. Panel E reports summary statistics across all university-field combinations with non-zero predictions within each scheme.

Across all schemes, "Elite National Universities" (Panel B) and "Top-Ranked Private Universities" (Panel C) exhibit the highest predicted match probabilities. For instance, Universidad Nacional de Córdoba shows a robust match probability of 77.23% for Engineering under the GPT-4o scheme. Engineering generally shows higher or more stable

⁵For instance, a university dummy may have high importance because it effectively separates high-match and low-match graduates, even if the university’s average effect on match probability is zero.

match probabilities than Psychology in several top-tier institutions. For example, at Universidad Nacional de Rosario, the probability drops from 76.35% (Eng.) to 72.14% (Psych.). This reinforces the idea that "specialist" fields suffer less frequent mismatch than "generalist" fields.

Notice that GPT-4.1 tends to assign much higher probabilities across the board (e.g., 70-75% for National Universities) compared to the more conservative Manual and GPT-4o models. While GPT-4.1 captures more institutions (lower LASSO penalty), it might be "over-smoothing" the prestige signal. The data shows a clear hierarchy. "Regional National Universities" (Panel C) like San Luis show significantly lower match probabilities (27.90% to 47.06%) compared to their "Elite" counterparts.

5 Conclusion and Discussion

At the critical juncture of academic selection, agents face a substantial ex-ante information deficit regarding the real returns and labor market insertion probabilities associated with specific disciplines. This lack of transparency leads to suboptimal equilibria, where a significant proportion of the labor force is relegated to sectors that are either misaligned with their training or characterized by the underutilization of acquired skills.

For the classification of education-job mismatch, we implement a hybrid labeling strategy of manual annotation and Large Language Model (LLM) assistance. Specifically, we utilize the advanced reasoning capabilities of GPT-4o and GPT-o1 to analyze the semantic alignment between educational backgrounds and job titles. Then, we compare the subsequent results with those derived from the manual labeling classification.

To assess the education-job mismatch phenomenon, we conduct a two-stage machine learning design that exploits high-dimensional educational and professional data collected from LinkedIn profiles. The first stage implements a Lasso model that identifies which institutional categories are statistically distinguishable from the baseline, while the second stage, develops a XGBoost technique to quantify their predictive importance through gain-based feature rankings. The feature importance analysis serves dual purposes in our framework. First, it validates the variable selection performed by Lasso by confirming that the selected variables contribute meaningfully to prediction. Second, it identifies the institutional categories that exhibit the strongest differential signaling effects, thereby informing the economic interpretation discussed in this paper.

The predicted probabilities shown here, serve as the empirical foundation for our signaling analysis, offering actionable information for prospective students and educational institutions regarding the expected labor market alignment associated with different educational trajectories. By isolating the predictive contribution of university and field of study, we provide a granular assessment of how educational choices map onto employment outcomes in the Argentine labor market. The 'Universidad Nacional de Córdoba' and 'Universidad Nacional de Rosario' appears as the "Gold Standard" signals in the GPT-4o model, as they maintain high probabilities for both fields. Conversely, the fact that under the Manual-labeling classification the model only retained a few private institutions suggests that the market may rely on a very narrow set of "elite" signals to guarantee a match. This analysis contributes to the broader literature on human capital signaling and educational returns by leveraging machine learning methods to uncover heterogeneous effects that would be difficult to detect in conventional regression frameworks.

The main limitation of this study lies in the restricted size of the dataset, both in terms of the overall sample and the number of observations available per university-major combination. The limited number of observations inevitably increases the variance of the estimates and may introduce bias in the measurement of institutional differences

in job–education match outcomes. This constraint stems from the substantial practical difficulties associated with collecting large-scale data through web scraping of LinkedIn profiles, given the platform’s security protocols and compliance safeguards.

Although the profiles selection was conducted in a random-picking manner, we cannot ensure the absence of selection bias since users on such platforms tend to be more digitally literate and exhibit higher levels of unobserved ability. Given that the purpose of this study is to focus on predicting education-job mismatch, the results are still valid but should be interpreted within this context. Moreover, the dataset does not include potentially relevant covariates such as academic performance, prior internships, socioeconomic background, or even relevant emotional dimensions (Mishra et al., 2016). The absence of these factors may generate residual confounding, as part of the observed variation attributed to universities could reflect differences in student characteristics rather than institutional effects per se.

The saturation degree of the specific labor market is another relevant issue to consider. This is a non-trivial factor when studying the match in the first post-graduation job, especially in popular and ‘credentialed’ fields of study such as psychology (since it might force mismatches towards HR or administration). The results found in this work are also subject to this type of labor market condition.

As a result of the forementioned constraints, the findings should be interpreted as exploratory rather than definitive. Nevertheless, the primary contribution of this study lies in its conceptual framework, methodological application, and data construction process. The paper proposes a novel approach to measuring university–labor market match using professional profile data and develops a replicable empirical strategy that can be scaled in future research. Subsequent extensions should focus on assembling a substantially larger dataset, both by increasing the number of individual observations and by incorporating a broader range of universities and academic programs. Expanding the empirical scope in this way would enhance statistical power, allow for the inclusion of richer predictors, and generate more robust and generalizable conclusions. Grounded in the structural nuances of the Argentine labor market, we hope this study provides actionable insights that serve as an informational benchmark for both educational institutions and students.

APPENDIX

A1. Data

Prompt Used for the Matching Classification

Eres un investigador profesional. Considera la Carrera estudiada por cada individuo. Debes determinar si el Primer trabajo del individuo es acorde a la carrera estudiada en cuestión. Si esto se cumple, hay match. Si no se cumple, no hay match. Adicionalmente, el cargo debe ser acorde a la calificación del individuo. Si el trabajo se encuentra por debajo de la calificación de alguien con un título universitario, es decir, el individuo está sobrecalificado, no hay match.

Carrera: {carrera}

Primer trabajo: {primer_trabajo}

Empresa: {empresa}

Responde únicamente con:

1 si es match

0 si es mismatch

Number of Observations by Educational Area

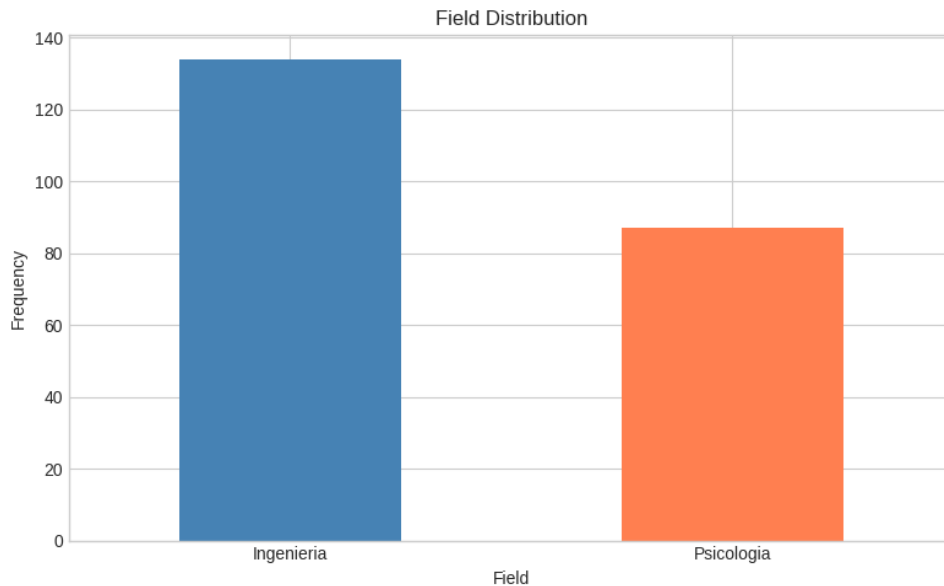


Figure A1: Number of observations by educational area.

A2. LASSO Results

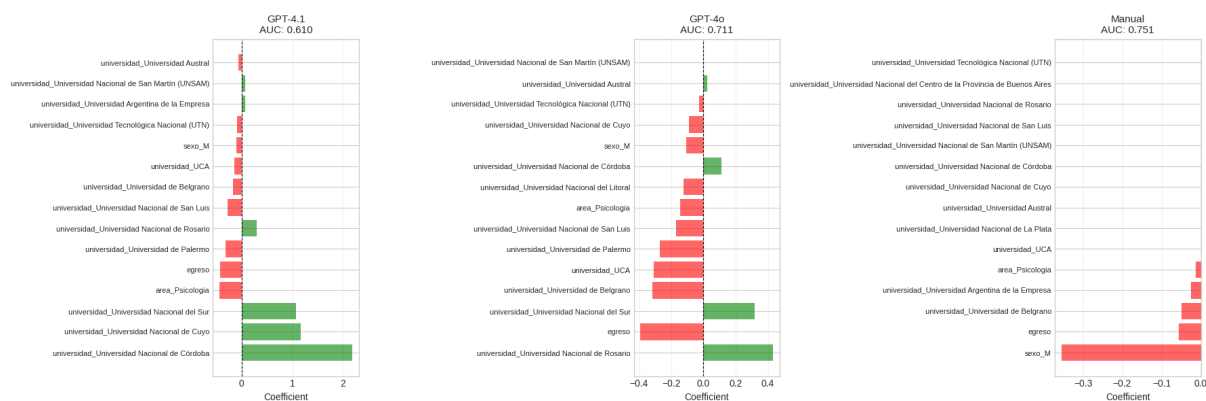


Figure A2: LASSO coefficients by matching strategy.

A3. Gradient Boosting Results

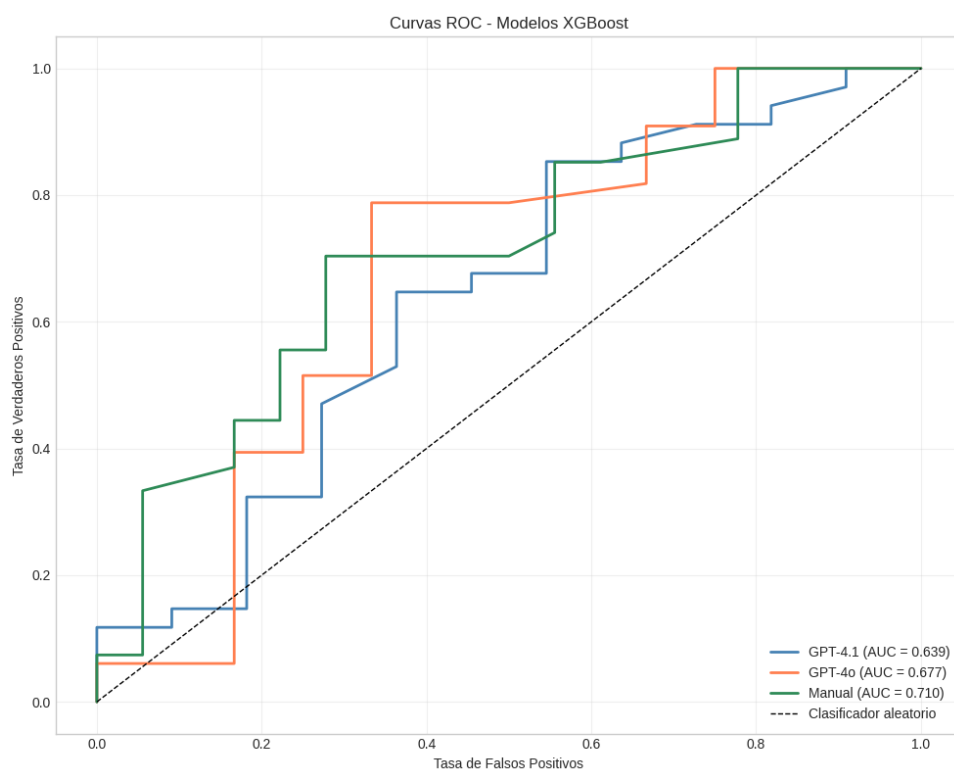


Figure A3: ROC Curve.

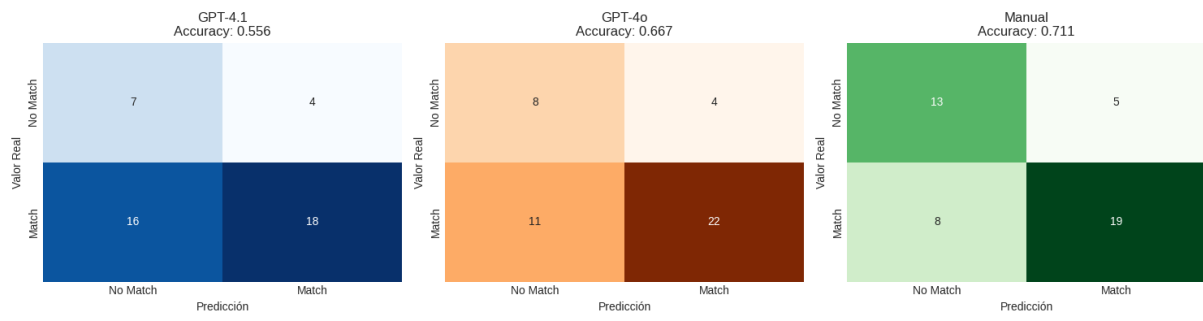


Figure A4: Confusion Matrices by Matching Strategy



Figure A5: Feature Importance

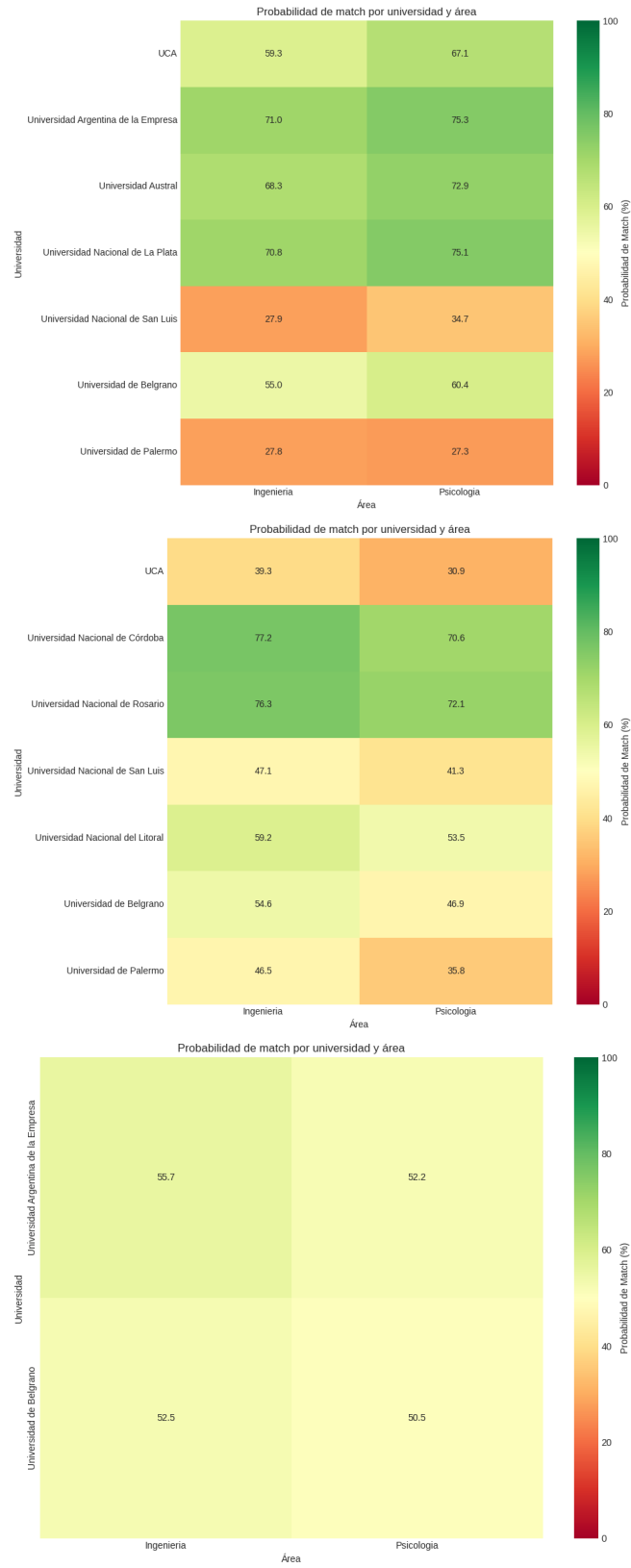


Figure A6: Distribution of job–education match probabilities across dimensions.

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